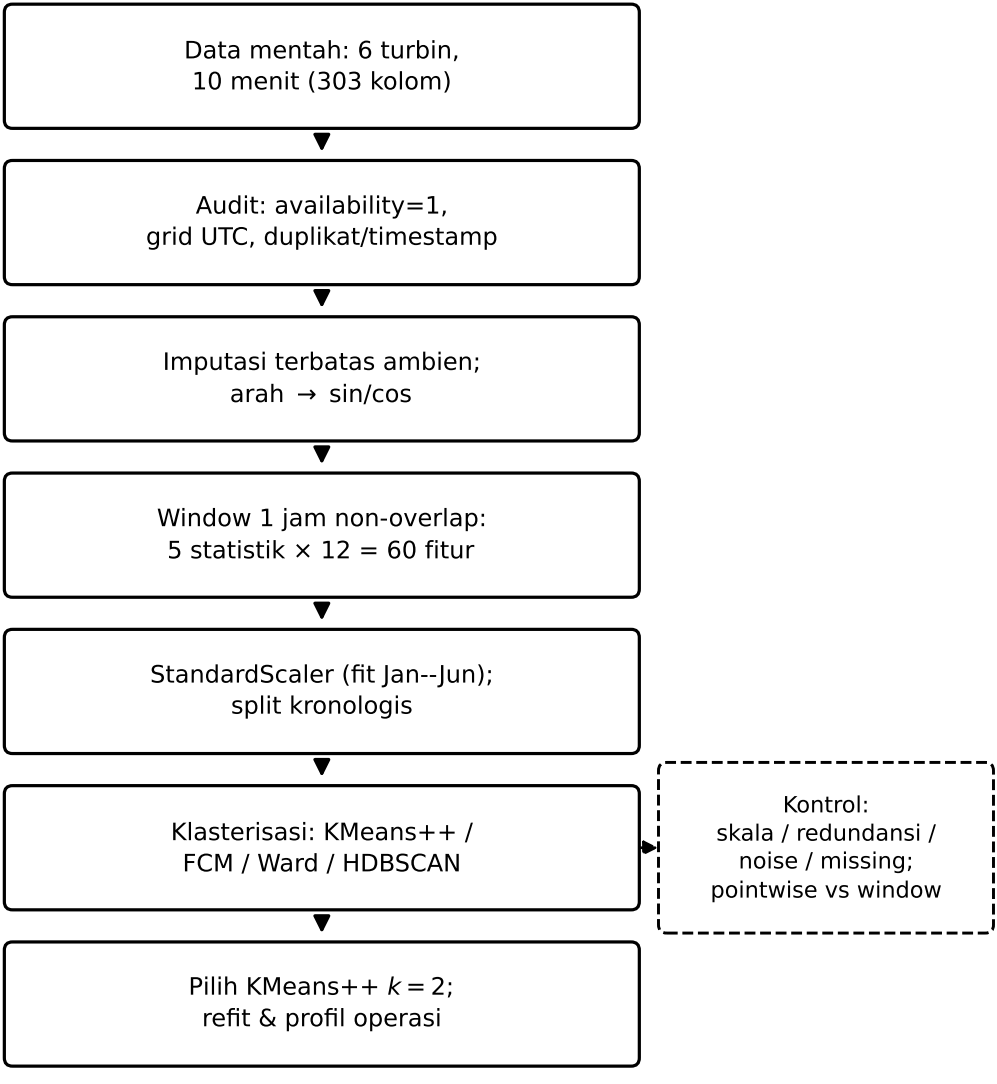
Data mentah: 6 turbin,
10 menit (303 kolom)
Audit: availability=1,
grid UTC, duplikat/timestamp
Imputasi terbatas ambien;
arah → sin/cos
Window 1 jam non-overlap:
5 statistik × 12 = 60 fitur
StandardScaler (fit Jan--Jun);
split kronologis
Klasterisasi: KMeans++ /
FCM / Ward / HDBSCAN
Kontrol:
skala / redundansi /
noise / missing;
pointwise vs window
Pilih KMeans++ $k = 2$;
refit & profil operasi